





Methodological guidelines for the generation of Linked Data

Data linking and publishing

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Based on the slides from Filip Radulovic, Paola Espinoza-Arias

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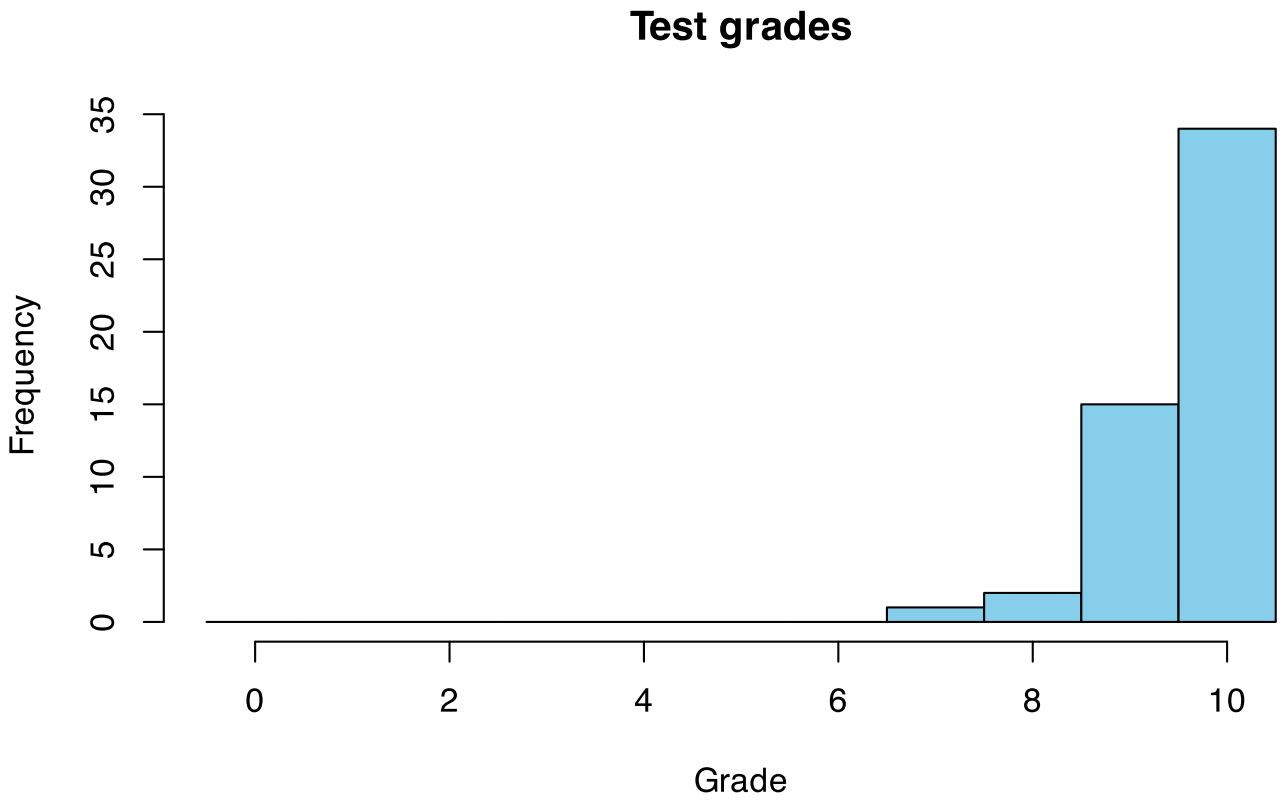
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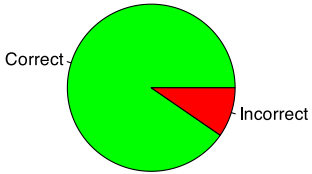
- To watch the video with a conversion of an Excel file to RDF with links to DBpedia and Europeana
  - http://youtu.be/1irwjiUOh_4
- To read about the different reconciliation services available for OpenRefine
  - https://github.com/OpenRefine/OpenRefine/wiki/Reconcilable-Data-Sources
- To read how reconciliation works in practice with Wikidata
  - https://github.com/OpenRefine/OpenRefine/wiki/Reconciliation
- To take pre-class test 5




Test grades
Frequency
0
5
10
15
20
25
30
35
0
2
4
6
8
10
Grade

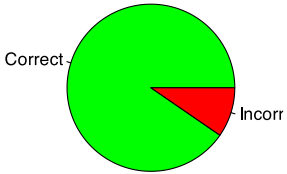
Pre-class test 5 - Questions

Q1



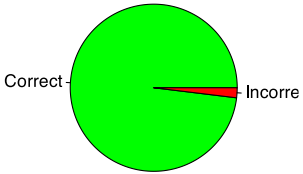
OpenRefine allows using a reconciliation task with a CSV file

Q2



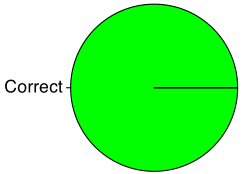
OpenRefine provides an API for reconciliation

Q3



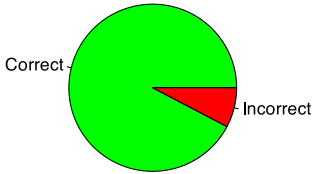
Reconciliation allows to identify resources in different datasets that refer to the same entity

Q4



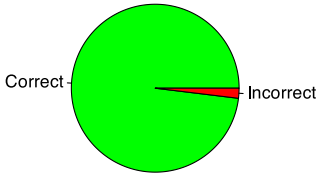
I can improve reconciliation if I restrict the matches according to the type of the resources

Q5



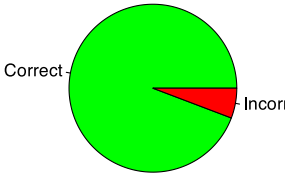
A resource that is defined with multiple properties will be more difficult to reconcile than a resource that is not defined by any property

Q6



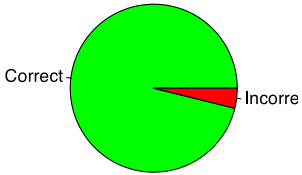
OpenRefine only allows to perform reconciliation with the English version of wikidata

Q7



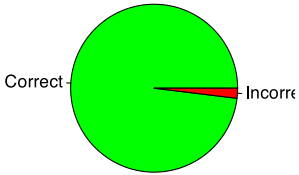
If I extend my dataset with wikidata data, I can use it for commercial purposes

Q8



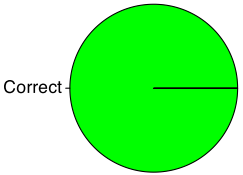
Which of the following OpenRefine reconciliation services allows reconciling with geographical data?

Q9



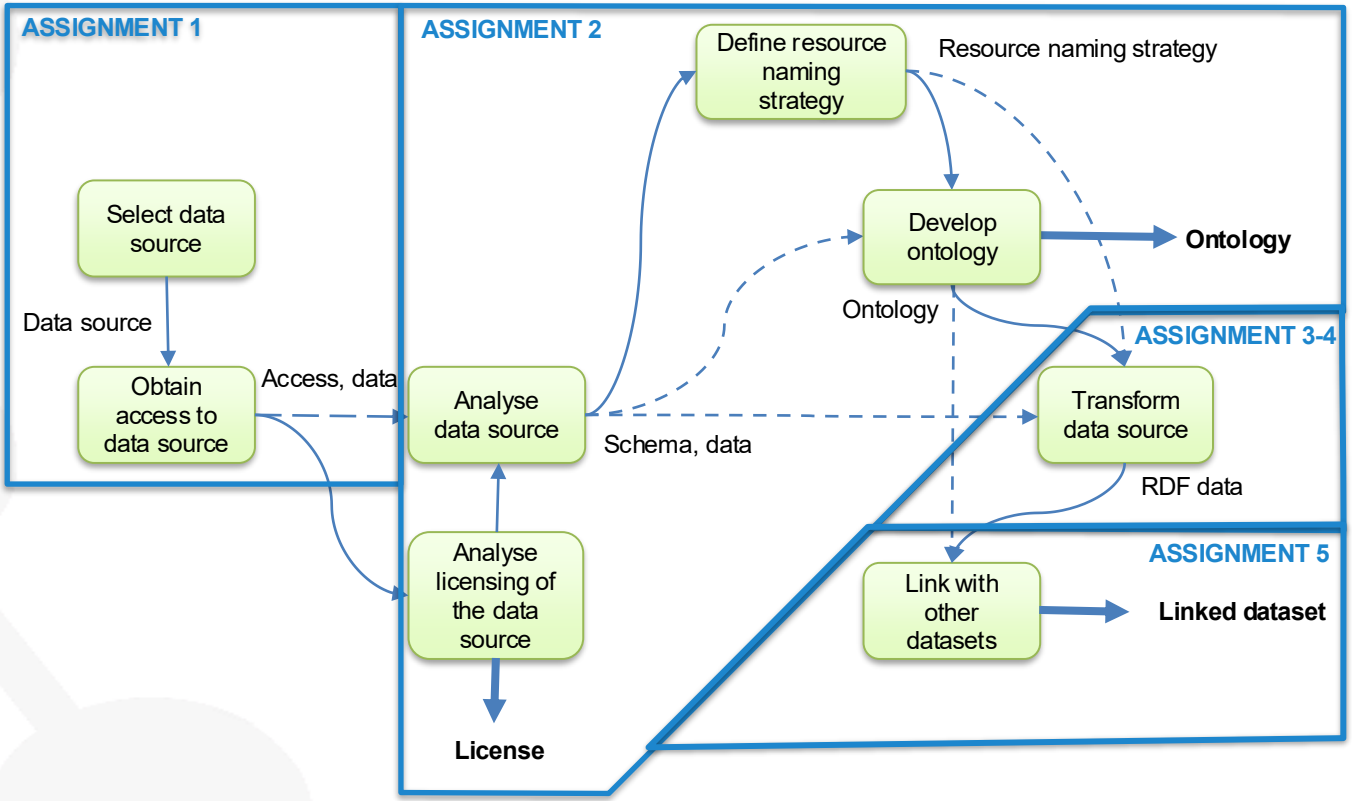
The reconciliation process in OpenRefine is fully automatic

Q10



In a museums dataset, I could reconcile the data from these museums with the data from the museums in DBpedia

Linked Data generation process



Deliverable D4.1: Requirements and guidelines for energy data generation. *Filip Radulovic, Raúl García-Castro, María Poveda-Villalón, Matthias Weise, Thanasis Tryferidis.* READY4SmartCities technical report. May 2014

Why should we link data?





Marina del Rey (Q988140)

census designated place

Statements

Own statements	From related entities
area	3.767749 square kilometre
population	8866 point in time : 2010
elevation above sea level	0 metre
located in time zone	Pacific Time Zone (time zone in North America)
	UTC−07:00 (Identifier for a time offset from UTC of −7) valid in period : daylight saving time (practice of advancing clocks so that evenings have more daylight and mornings have less)
	UTC−08:00 (Identifier for a time offset from UTC of −8) valid in period : standard time (synchronization of clocks within a geographical area or region)
coordinate location	(33°58'45" N , 118°27'10" W)
located in the administrative territorial entity	Los Angeles County (county in California, United States of America)
country	United States of America (federal republic in North America)

Media

page banner	Marina del Rey banner.jpg
locator map image	LA County Incorporated Areas Marina del Rey highlighted.svg
Commons category	Marina del Rey, California
image	Marina Del Rey Looking South.jpg

Wikimedia Categories and Portals

Own statements	From related entities
category of associated people	Category:People from Marina del Rey, California (Wikimedia category)
category for people born here	Q21000701 (Wikimedia category)
category for people who died here	Q21000703 (Wikimedia category)
topic's main category	Category:Marina del Rey, California (Wikimedia category)

Links

Wikidata page	
Wikipedia article	
Reasonator	

Identifiers

MusicBrainz area ID	2a24c2...3f0fdd
archINFORM location ID	16300
DMOZ ID	Region...l_Rey/
FIPS 55-3 (locations in the US)	06-45806
WOEID	2445713
Who's on First ID	85928367
Freebase ID	/m/01zhsqk
Library of Congress authority ID	n81081018
VIAF ID	247815515
GND ID	4418951-5
GNIS ID	1852255
GeoNames ID	5370542

WIKIDATA

- Find equivalents in other datasets
- Ensuring data are not isolated
- Easing up data integration and interoperability
- Incorporating additional information from existing sources
  - Locations
  - Pictures
  - Metadata
  - etc.

Steps

1. To identify classes whose instances can be linked
2. To identify data sets that may contain instances for the previously-identified classes
3. To select the tool(s) for performing the task
4. To use the tool(s) in order to obtain links

owl:sameAs

http://dbpedia.org/resource/Barack_Obama ⟷ owl:sameAs ⟷ https://www.wikidata.org/wiki/Q76





Matching categories

- **Value matching**. Determining whether two values of two properties expressed in different ways are equivalent
  - **A value match of two properties of two instances does not always denote an equivalence**
  - Based on: string similarity, keyword-based search (e.g., using Google), semantic relations between the values (e.g., using WordNet), etc.
- **Individual matching**. Determining whether two instances are equivalent
  - According to the descriptions of these instances
  - Based on: aggregation of value matching techniques, the transitivity of *owl:sameAs*, ontology mappings, etc.
- **Dataset matching**. Utilizing a set of potential individual mappings between two datasets
  - Datasets are analyzed as a whole

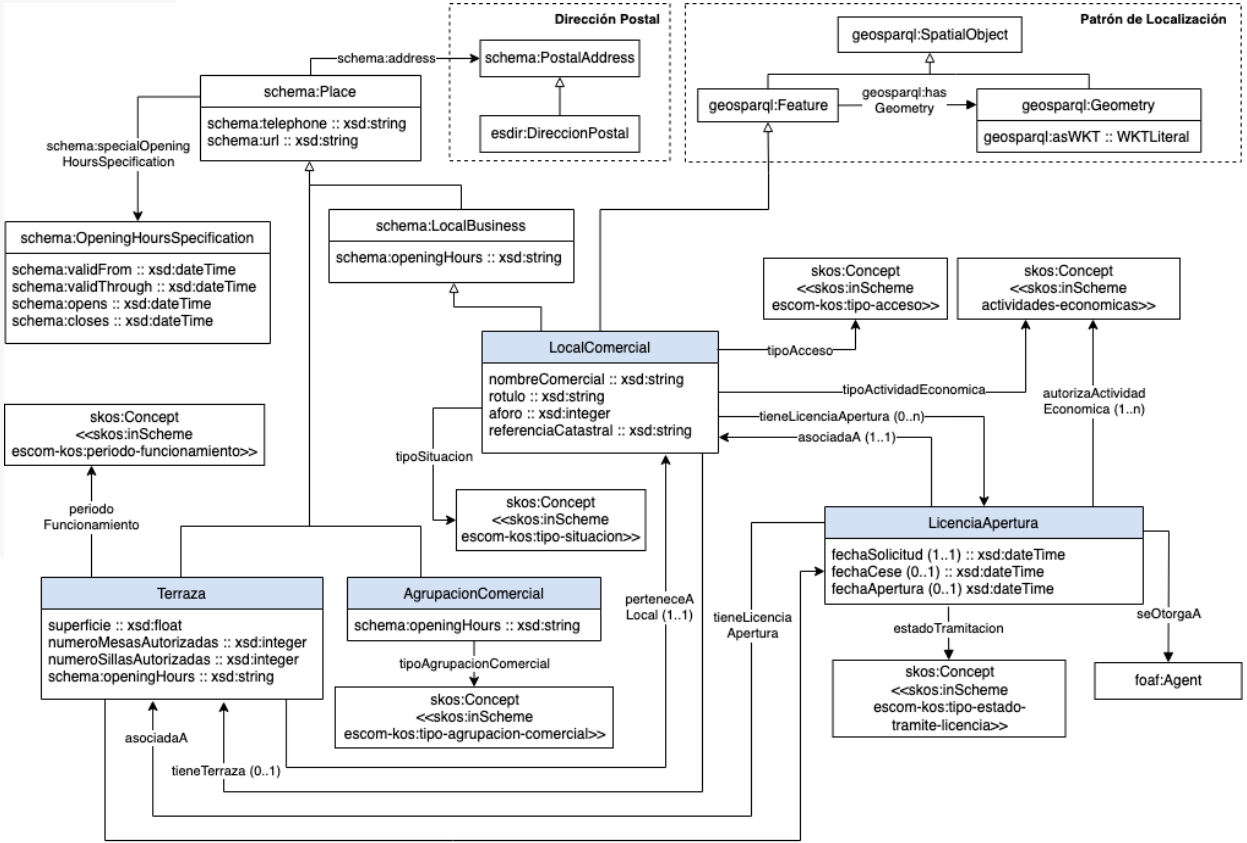
Linking technologies

Tool	Description	Matching type
LD mapper	Dataset linking tool based on Prolog that uses a similarity aggregation algorithm	Value, individual
LIMES	Semi-automatic interlinking tool available both as a web service and through a user interface.	Value
LN2R	Instance matching system that consists of logical and numerical matching engines	Value, individual, dataset
RDF-AI	Tool that produces a set of links between datasets	Value, individual, dataset
Serimi	Tool that uses a two-phase approach according to information retrieval strategies and the examination of candidate descriptions	Value, dataset
Silk	Tool for interlinking datasets and maintaining the links obtained	Value, individual



M. Nentwig et al. “A survey of current Link Discovery frameworks”. Semantic Web, 8(3): 419-436, 2017, http://www.semantic-web-journal.net/system/files/swj1117.pdf

http://vocab.ciudadesabiertas.es/def/comercio/tejido-comercial/index-es.html



Ontology for describing Local businesses

https://github.com/opencitydata/guia-rdf-datosgob

id_terraza	id_local	id_distrito	local	desc_distrito_local	id_barrio_local	desc_barrio_local	id_ndp_edificio	id_clase_ndp_edificio	id_vial_edificio	clase_vial_edificio	desc_vial_edificio	nom_edificio
2706	280009803		1	CENTRO	104	JUSTICIA	11004774	1	670700	CALLE	SAN GREGORIO	NUM
2707	280007810		1	CENTRO	101	PALACIO	11001071	1	797800	TRAVESIA	VISTILLAS	NUM
2708	285001360		1	CENTRO	101	PALACIO	11000624	1	542500	CALLE	NUNCIO	NUM
2711	270041313		1	CENTRO	102	EMBAJADORES	11002341	1	71000	CALLE	ARGUMOSA	NUM
2712	285002923		1	CENTRO	104	JUSTICIA	11004218	1	207300	CALLE	CONDE DE XIQUENA	NUM
2713	270224434		1	CENTRO	102	EMBAJADORES	11003171	1	81600	CALLE	ATOCHA	NUM
2714	280010456		1	CENTRO	105	UNIVERSIDAD	11005645	1	313100	CALLE	FUENCARRAL	NUM
2715	280010388		1	CENTRO	105	UNIVERSIDAD	11005486	1	313100	CALLE	FUENCARRAL	NUM

- Classes with linkable instances: Distrito
- Data sets to link to: Wikidata
- Tool to use: OpenRefine

http://vocab.ciudadesabiertas.es/def/comercio/tejido-comercial/index-es.html

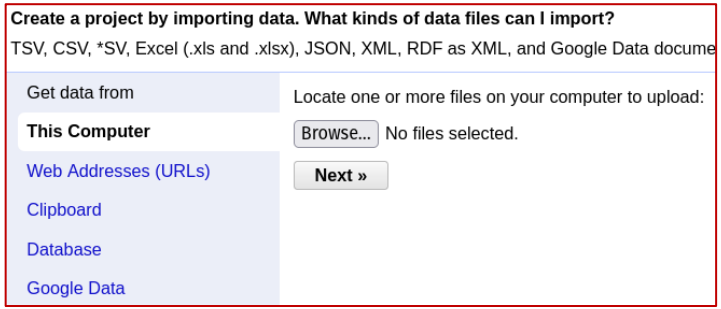
Goal:

<http://vocab.ciudadesabiertas.es/recurso/local-comercial/10003602> a **escom:LocalComercial** ;
escom:tieneLicenciaApertura <http://vocab.ciudadesabiertas.es/recurso/licencia-apertura/500-2017-11358> ;
dc:identifier "10003602"^^xsd:string ;
escom:rotulo "Restaurante Cocina del Norte"^^xsd:string ;
escom:aforo "30"^^xsd:int ; escom:referenciaCatastral "9951803" ;
schema:telephone "640439521"^^xsd:string ;
escom:tipoSituacion <http://vocab.linkeddata.es/datosabiertos/kos/comercio/tipo-situacion/activo> ;
escom:tipoActividadEconomica <http://vocab.linkeddata.es/datosabiertos/kos/comercio/cnae/Cod4/5610> ;
escom:tieneTerraza <http://vocab.ciudadesabiertas.es/recurso/terraza/10003602> ;
escom:tipoAcceso <http://vocab.linkeddata.es/datosabiertos/kos/comercio/tipo-acceso/puerta-calle> ;
geosparql:hasGeometry <http://vocab.ciudadesabiertas.es/recurso/local-comercial/10003602/geometry> ;
esadm:distrito <http://vocab.ciudadesabiertas.es/recurso/distrito/20> .
<http://vocab.ciudadesabiertas.es/recurso/distrito/20> a esadm:Distrito ;
owl:sameAs <http://www.wikidata.org/entity/Q1763376> ;
rdfs:label "CENTRO" .

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Selecting a column to reconcile

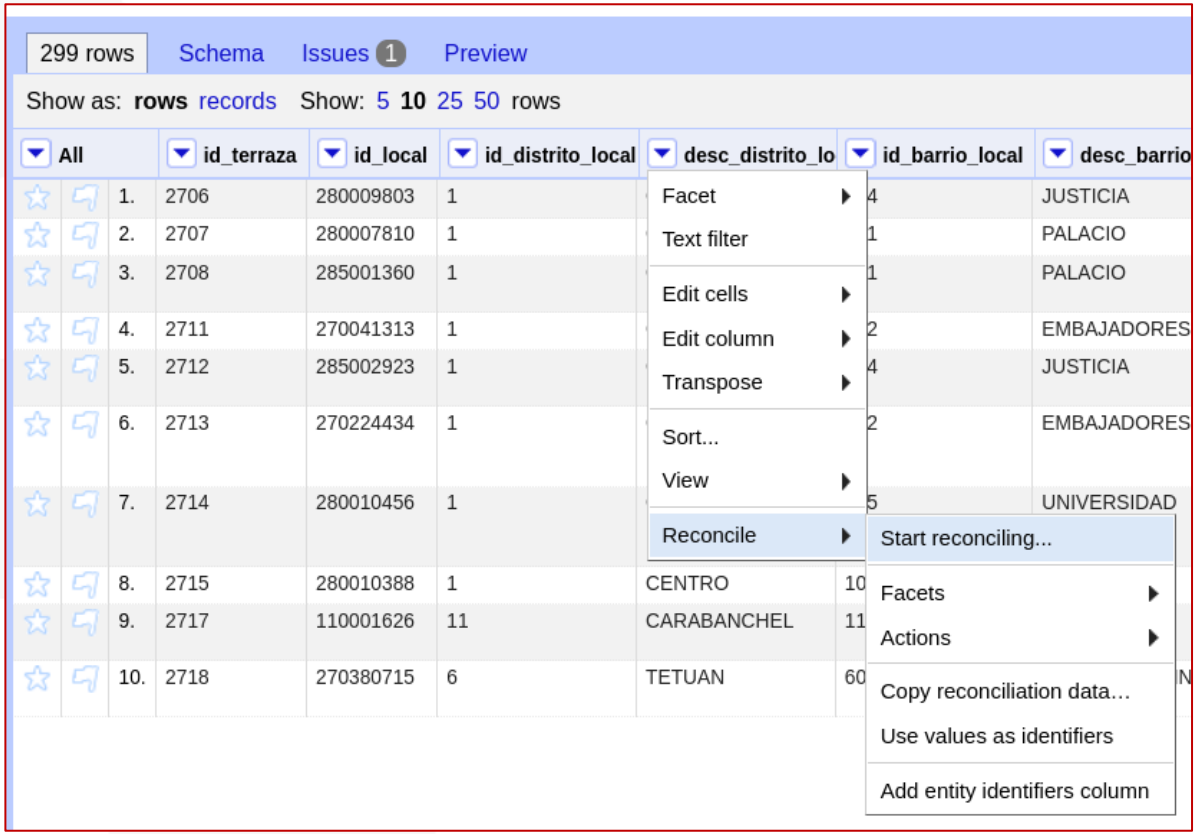


299 rows | Schema | Issues 1 | Preview | Extensions: Wikidata

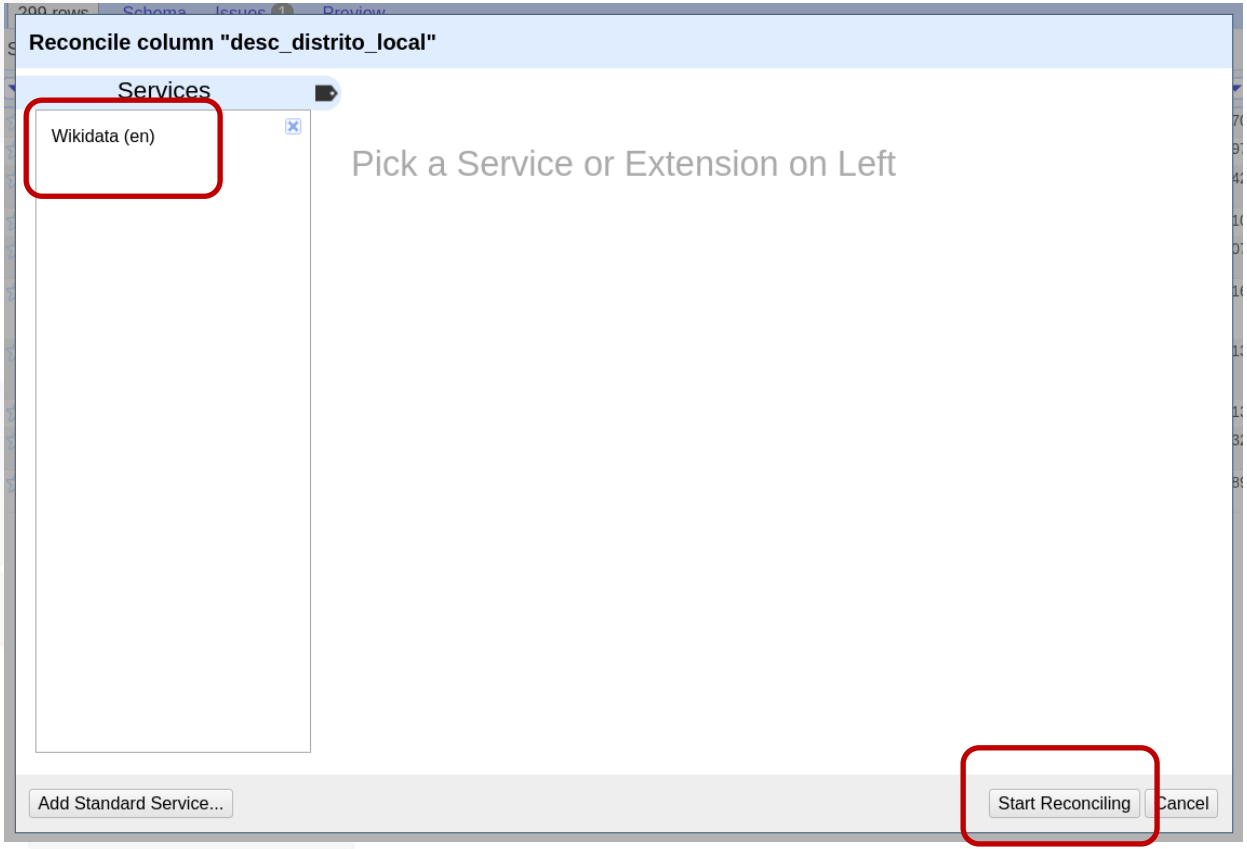
Show as: **rows** records | Show: 5 **10** 25 50 rows | « first ‹ previous **1 - 10** next › last »

All		id_terraza	id_local	id_distrito_local	desc_distrito_lo	id_barrio_local	desc_barrio_loc	id_ndp_edificio	id_clase_ndp_ed	id_vial_edificio	clase_vial_edific	desc_vial_edifi
	1.	2706	280009803	1	CENTRO	104	JUSTICIA	11004774	1	670700	CALLE	SAN GREGORIO
	2.	2707	280007810	1	CENTRO	101	PALACIO	11001071	1	797800	TRAVESIA	VISTILLAS
	3.	2708	285001360	1	CENTRO	101	PALACIO	11000624	1	542500	CALLE	NUNCIO
	4.	2711	270041313	1	CENTRO	102	EMBAJADORES	11002341	1	71000	CALLE	ARGUMOSA
	5.	2712	285002923	1	CENTRO	104	JUSTICIA	11004218	1	207300	CALLE	CONDE DE XIQUENA
	6.	2713	270224434	1	CENTRO	102	EMBAJADORES	11003171	1	81600	CALLE	ATOCHA
	7.	2714	280010456	1	CENTRO	105	UNIVERSIDAD	11005645	1	313100	CALLE	FUENCARRAL
	8.	2715	280010388	1	CENTRO	105	UNIVERSIDAD	11005486	1	313100	CALLE	FUENCARRAL
	9.	2717	110001626	11	CARABANCHEL	1102	OPAÑEL	11070474	1	332700	CALLE	GENERAL RICARDOS
	10.	2718	270380715	6	TETUAN	602	CUATRO CAMINOS	11029335	1	88906676	CALLE	EDGAR NEVILLE

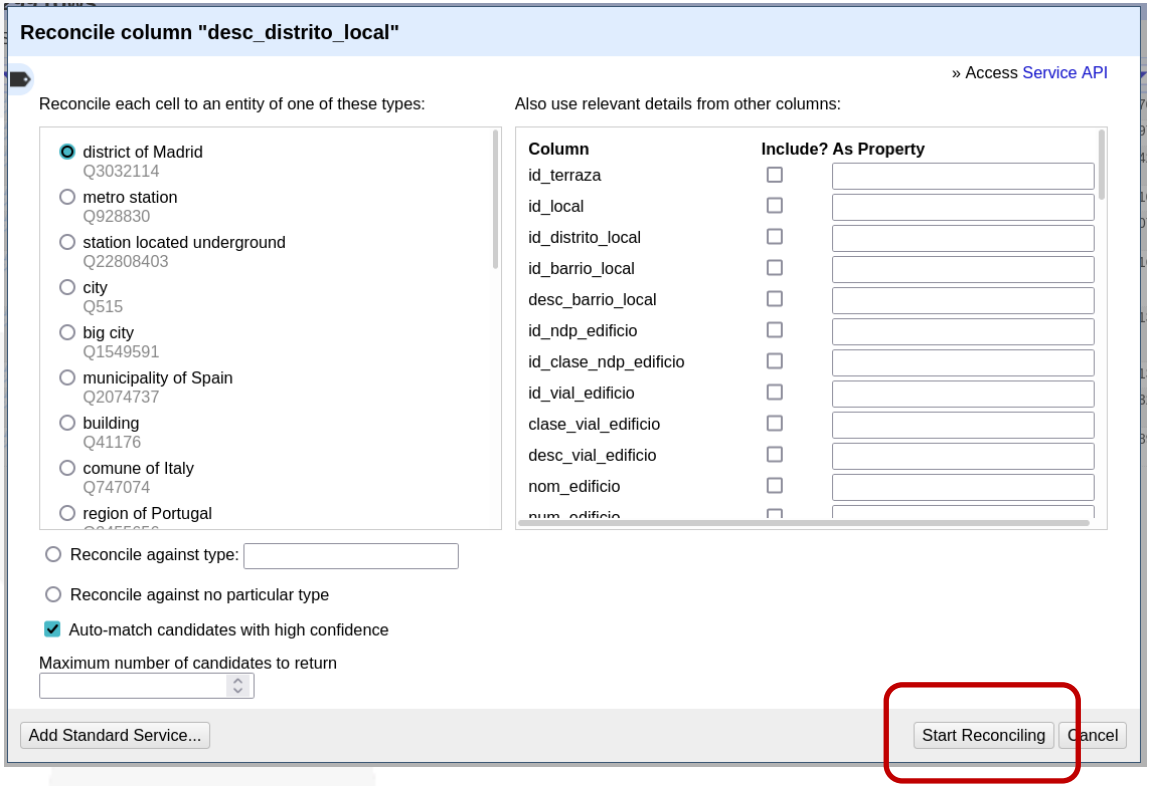
Reconciling values



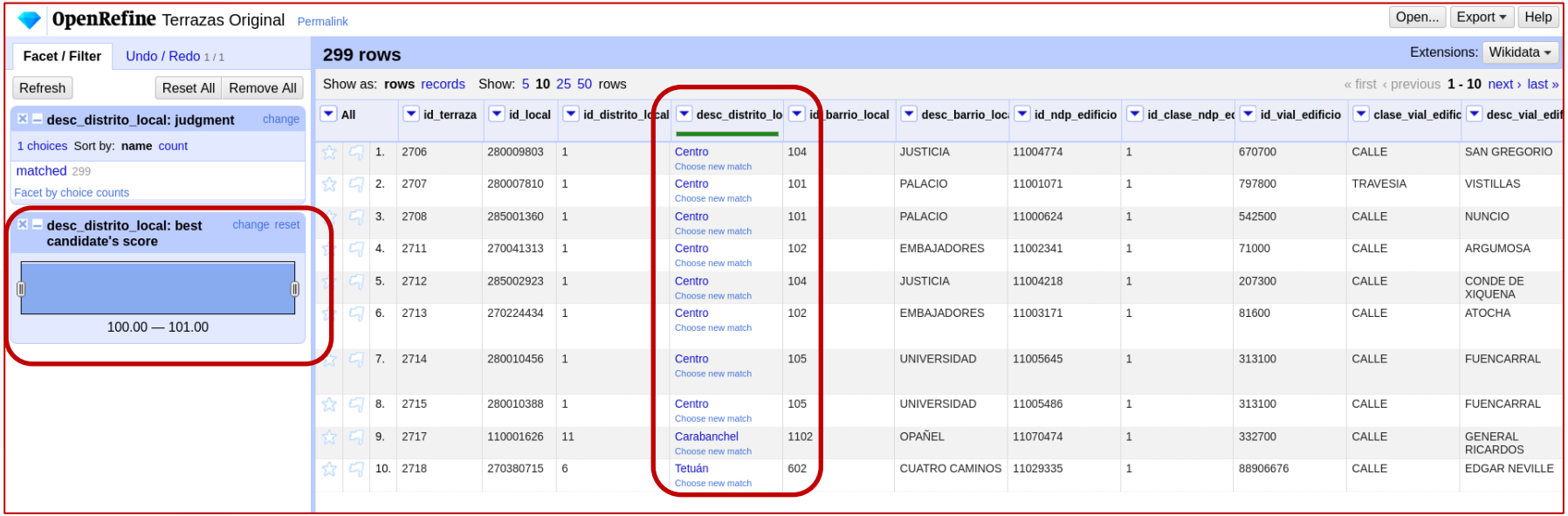
Reconciling values



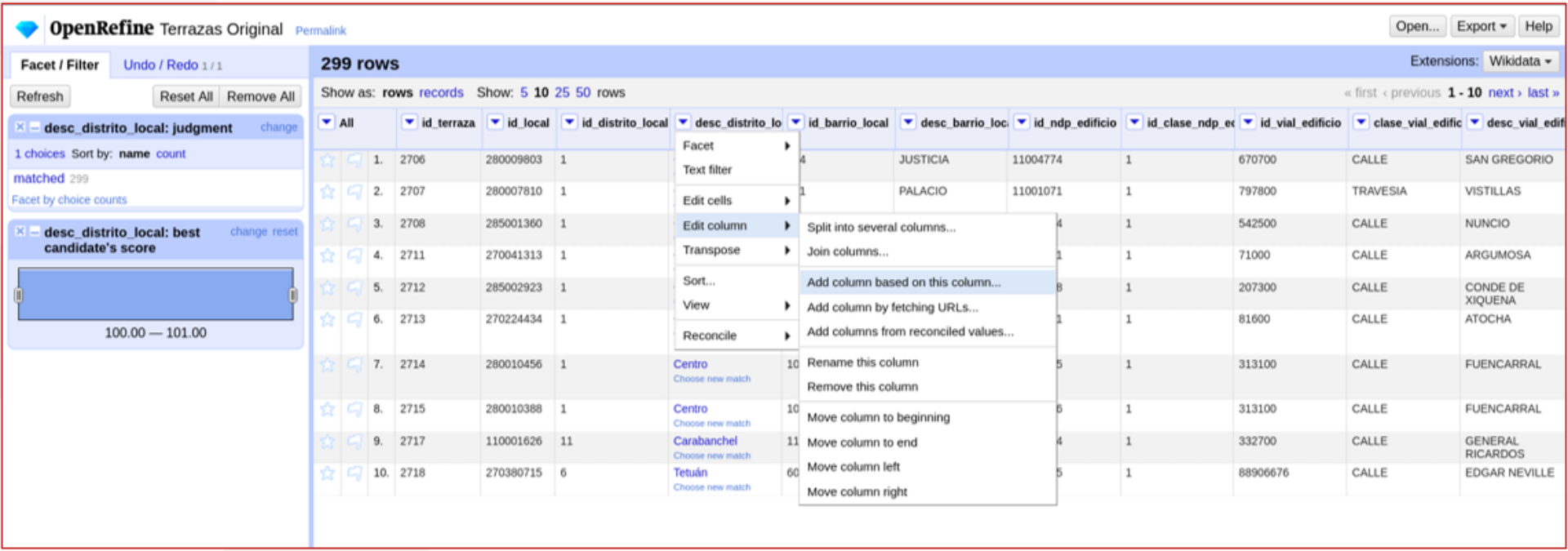
Reconciling values



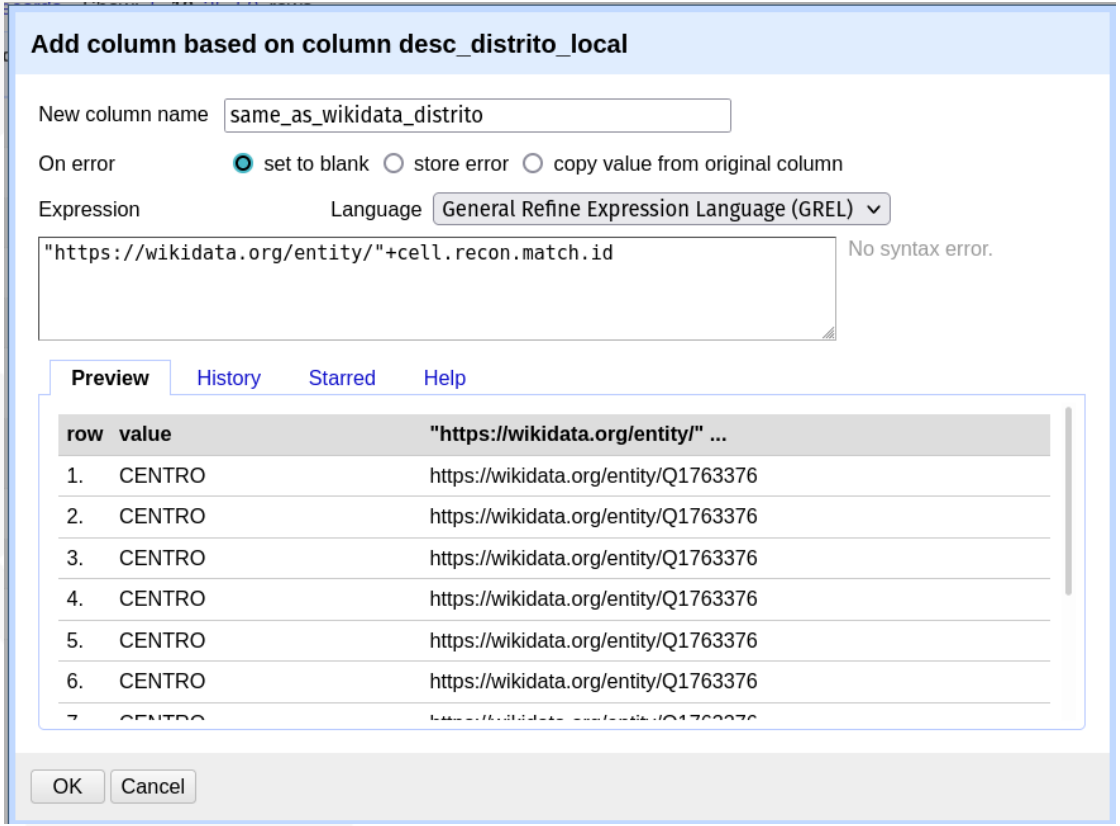
Reconciling values



Reconciling values



Separating columns



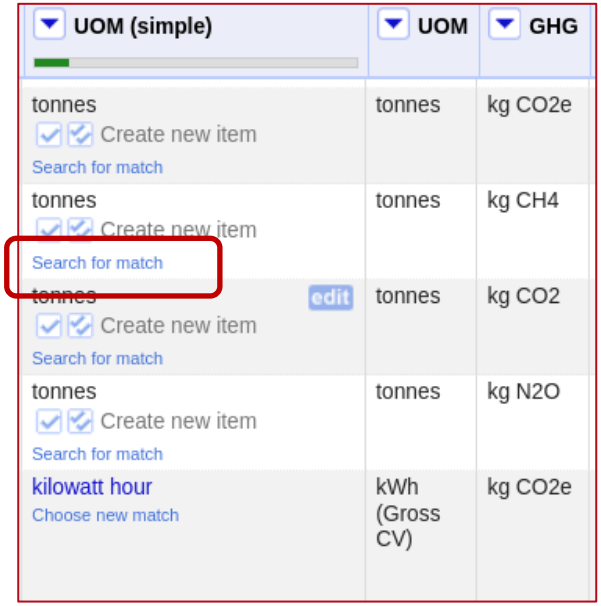
Result

Show as: **rows** records Show: 5 **10** 25 50 rows « first ‹ previous **1 - 10** next › last »

All		id_terraza	id_local	id_distrito_local	desc_distrito_lo	same_as_wikida	id_barrio_local	desc_barrio_loc	id_ndp_edificio	id_clase_ndp_ed	id_vial_edificio	clase_vial_edifi
1.	2706	280009803	1	Centro Choose new match	https://wikidata.org/entity/Q1763376	104	JUSTICIA	11004774	1	670700	CALLE
2.	2707	280007810	1	Centro Choose new match	https://wikidata.org/entity/Q1763376	101	PALACIO	11001071	1	797800	TRAVESIA
3.	2708	285001360	1	Centro Choose new match	https://wikidata.org/entity/Q1763376	101	PALACIO	11000624	1	542500	CALLE
4.	2711	270041313	1	Centro Choose new match	https://wikidata.org/entity/Q1763376	102	EMBAJADORES	11002341	1	71000	CALLE
5.	2712	285002923	1	Centro Choose new match	https://wikidata.org/entity/Q1763376	104	JUSTICIA	11004218	1	207300	CALLE
6.	2713	270224434	1	Centro Choose new match	https://wikidata.org/entity/Q1763376	102	EMBAJADORES	11003171	1	81600	CALLE
7.	2714	280010456	1	Centro Choose new match	https://wikidata.org/entity/Q1763376	105	UNIVERSIDAD	11005645	1	313100	CALLE
8.	2715	280010388	1	Centro Choose new match	https://wikidata.org/entity/Q1763376	105	UNIVERSIDAD	11005486	1	313100	CALLE
9.	2717	110001626	11	Carabanchel Choose new match	https://wikidata.org/entity/Q1001991	1102	OPAÑEL	11070474	1	332700	CALLE
10.	2718	270380715	6	Tetuán Choose new match	https://wikidata.org/entity/Q1773540	602	CUATRO CAMINOS	11029335	1	88906676	CALLE

What if there is no initial match?

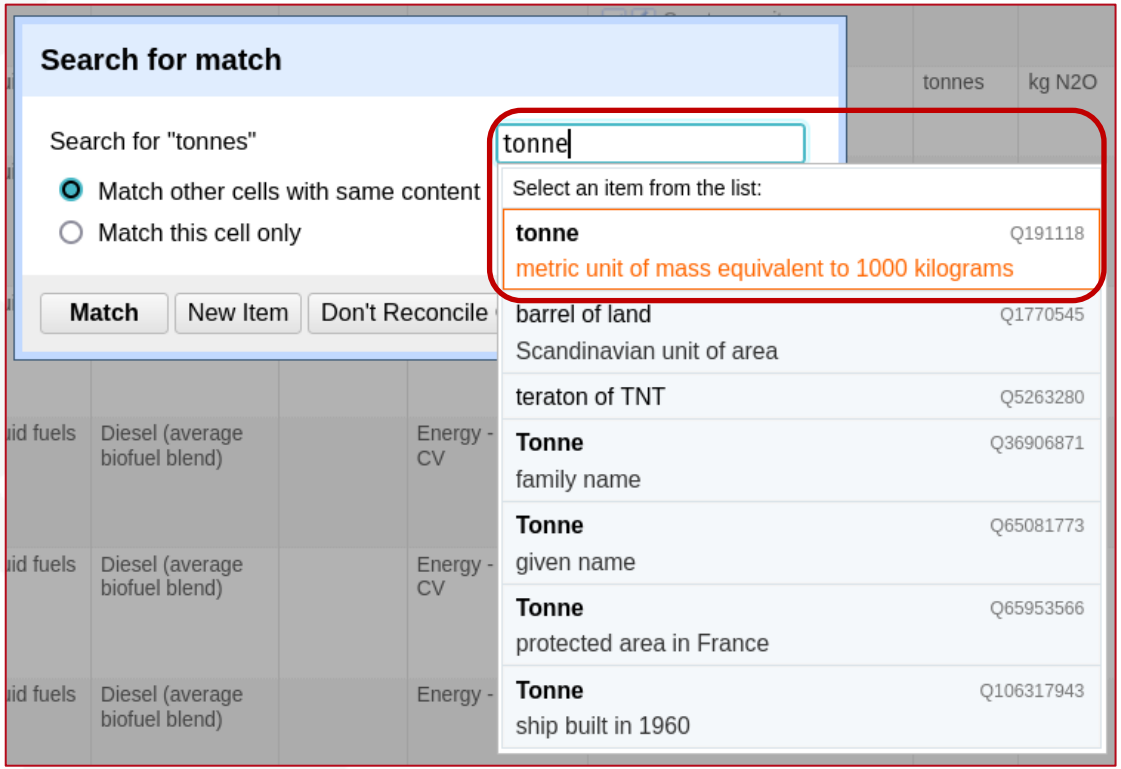
You can search suggestions (note: different example table!):



UOM (simple)	UOM	GHG
tonnes Create new item Search for match	tonnes	kg CO2e
tonnes Create new item Search for match	tonnes	kg CH4
tonnes edit Create new item Search for match	tonnes	kg CO2
tonnes Create new item Search for match	tonnes	kg N2O
kilowatt hour Choose new match	kWh (Gross CV)	kg CO2e

What if there is no initial match?

You can search suggestions:



results.csv

id_distrito_local	same_as_wikidata_distrito
1	https://wikidata.org/entity/Q1763376

```
@prefix rr: <http://www.w3.org/ns/r2rml#>.
@prefix rml: <http://semweb.mmlab.be/ns/rml#>.
@prefix esadm: <http://vocab.linkeddata.es/datosabiertos/def/sector-publico/territorio#/> .
@prefix rdfs: <http://www.w3.org/2000/01/rdf-schema#> .
@prefix owl: <http://www.w3.org/2002/07/owl#>.

<#TriplesMap1>
  rr:logicalTable [ rml:logicalSource "results.csv" ];
  rr:subjectMap [
    rr:template "http://example.org/Distrito/{id_distrito_local}";
    rr:termType rr:IRI; rr:class esadm:Distrito;
  ];
  rr:predicateObjectMap [
    rr:predicateMap [ rr:constant rdfs:label; rr:termType rr:IRI ]
    rr:objectMap [ rml:reference "desc_distrito_local"; rr:termType rr:Literal ]
  ];
  rr:predicateObjectMap [
    rr:predicateMap [ rr:constant owl:sameAs; rr:termType rr:IRI ]
    rr:objectMap [
       rr:template "same_as_wikidata_distrito";
       rr:termType rr:IRI ]
         ];
          …
         .
```




```
<http://example.org/Distrito/1> a esadm:Distrito;
            rdfs:label "CENTRO" ;
            owl:sameAs <http://www.wikidata.org/entity/Q1763376> .
```

results.csv

id_distrito_local	same_as_wikidata_distrito
1	https://wikidata.org/entity/Q1763376

```
prefixes:
  ex-resource: "https://example.org"
  owl: "http://www.w3.org/2002/07/owl#"
  schema: "https:schema.org/"
  esadm: "http://vocab.linkeddata.es/datosabiertos/def/sector-publico/territorio#"

mappings:
  places:
    sources:
      - ['results.csv~csv']
    s: ex-resource:Distrito/$(id_distrito_local)
    po:
      - [a, esasdm:Distrito]
      - [rdfs:label, $(desc_distrito_local)]
      - [owl:sameAs, $(same_as_wikidata_distrito)~iri]
...
```



```
<http://example.org/Distrito/1> a esadm:Distrito;
            rdfs:label "CENTRO" ;
            owl:sameAs <http://www.wikidata.org/entity/Q1763376> .
```

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Describing datasets with metadata

- Consists of providing (machine-readable) metadata of RDF datasets which can be processed by engines
- This information allows for:
  - Efficient and effective search of datasets
  - Selection of appropriate datasets (for consumption or interlinking)
  - Get general statistics of the datasets
- Main vocabularies for metadata description:
  - **VoID (Vocabulary of Interlinked Datasets)**
  - **DCAT (Data Catalog Vocabulary)**

Vocabulary of Interlinked Datasets (VoID)

- https://www.w3.org/TR/void
  - W3C Interest Group Note from 2011
  - The fundamental concept is the RDF dataset represented by the concept `void:Dataset`
  - Covers 4 types of metadata:
    1. General metadata
    2. Structural metadata
    3. Access metadata
    4. Descriptions of linksets
  - Provides guidelines for deployment and discovery

VoID: General metadata (I)

- Used by potential users to identify appropriate datasets
- Provides information about the title of the dataset, relevant dates, contact person/organization, data subject, among other technical features
- Reuses predicates from well-known vocabularies such as Dublin Core Metadata Terms and FOAF

VoID: General metadata (II)

- General information:

```
:DBpedia a void:Dataset ;
    dcterms:title "DBPedia";
    dcterms:description "RDF data extracted
    from Wikipedia";
    foaf:homepage <http://dbpedia.org/>;
    dcterms:contributor :FU_Berlin;
    dcterms:contributor :University_Leipzig;
    dcterms:contributor :OpenLink_Software;
    dcterms:contributor :DBpedia_community;
    dcterms:source
    <http://dbpedia.org/resource/Wikipedia>;
    dcterms:modified "2008-11-17"^^xsd:date.
```

Predicate	Range	Description
dcterms:title	Literal	Name of the dataset.
dcterms:description	Literal	Description of the dataset.
dcterms:source	RDF resource	Source from which the dataset was derived.
dcterms:creator	RDF resource	Primarily responsible of creating the dataset.
dcterms:date	xsd:date	Time associated with an event in the life-cycle of the resource.
dcterms:created	xsd:date	Date of creation of the dataset.
dcterms:issued	xsd:date	Date of publication of the dataset.
dcterms:modified	xsd:date	Date on which the dataset was changed.
foaf:homepage	Literal	Homepage of the dataset.
dcterms:publisher	RDF resource	Entity responsible for making the dataset available.
dcterms:contributor	RDF resource	Entity responsible for making contributions to the dataset.

Source: https://www.w3.org/TR/void/#metadata

VoID: General metadata (III)

- Other information:
  - License of the dataset (`dcterms:license`): specifies the usage conditions of the data
  - Category of the dataset (`dcterms:subject`): specifies the topics or domains covered by the dataset
  - Technical features (`void:feature`): express technical properties of the data (e.g., RDF serialization formats)

```
:Geonames a void:Dataset ;
    dcterms:license <https://creativecommons.org/licenses/by/4.0/>;
    dcterms:subject <http://dbpedia.org/resource/Location>;
    void:feature <http://www.w3.org/ns/formats/RDF_XML>; .
```

VoID: Structural metadata (I)

- Provides high-level information about the schema and internal structure of a dataset
- Useful when exploring or querying datasets
- Includes information such as the vocabularies used in the dataset, statistics about the size of the dataset, and examples of typical resources in the dataset

VoID: Structural metadata (II)

- Information about resources:
  - Example resources (`void:exampleResource`): provides representative example entities for a dataset to allow users to get an impression of the kind of data that is present in the dataset
  - Pattern for resource URIs (`void:uriSpace`): states that all entity URIs in a dataset start with a given string

```
:DBpedia a void:Dataset ;
    void:exampleResource <http://dbpedia.org/resource/Berlin> ;
    void:exampleResource <http://dbpedia.org/resource/Physics> ;
    void:exampleResource <http://dbpedia.org/resource/Ludwig_van_Beethoven> ;
    void:uriSpace "http://dbpedia.org/resource/" .
```

VoID: Structural metadata (III)

- Vocabularies used in the dataset:
  - The `void:vocabulary` property can be used to list vocabularies or ontologies used in a dataset
  - Typically, only the most relevant vocabularies are listed
  - This property can only be used for entire vocabularies; it cannot be used to express that a subset of the vocabulary occurs in the dataset

```
:DBpedia a void:Dataset ;
    void:vocabulary <http://dbpedia.org/ontology/>.
```

VoID: Structural metadata (IV)

- Statistics about a dataset
  - Express numeric statistics about a dataset:

Predicate	Range	Description
void:triples	Number	Total number of triples contained in the dataset.
void:entities	Number	Total number of entities that are described in the dataset. An entity must have a URI, and match the void:uriRegexPattern
void:classes	Number	Total number of distinct classes in the dataset.
void:properties	Number	Total number of distinct properties in the dataset.
void:distinctSubjects	Number	Total number of distinct subjects in the dataset.
void:distinctObjects	Number	Total number of distinct objects in the dataset.
void:documents	Number	Total number of documents, in case that the dataset is published as a set of individual documents.

Source: https://www.w3.org/TR/void/#metadata

VoID: Structural metadata (V)

- Partitioned data sets
  - The `void:subset` property provides description of parts of a dataset

```
:DBpedia a void:Dataset;
    void:subset :DBpedia_shortabstracts;
    void:subset :DBpedia_infoboxes.
```

  - Datasets can be partitioned based on classes or properties:
    - `void:classPartition` contains only instances of a particular class
    - `void:propertyPartition` contains only triples with a particular predicate

```
:MyDataset a void:Dataset;
    void:classPartition [ void:class foaf:Person; ];
    void:classPartition [ void:class foaf:Organization; ];
    void:propertyPartition [ void:property foaf:name; ];
    void:propertyPartition [ void:property foaf:member; ];
    void:propertyPartition [ void:property foaf:homepage; ];
    void:propertyPartition [ void:property rdf:type; ].
```

VoID: Access metadata

- The access metadata describes the methods of accessing the actual RDF dataset

Method	Predicate	Description
URI look up endpoint	void:uriLookupEndpoint	Specifies the URI of a service for accessing the data set (different from the SPARQL protocol)
Root resource	void:rootResource	URI of the top concepts (only for data sets structured as trees)
SPARQL endpoint	void:sparqlEndpoint	Provides access to the data set via the SPARQL protocol*
RDF data dumps	void:dataDump	Specifies the location of the dump file. If the data set is split into multiple files, then several values of this property are provided

* This assumes that the default graph of the SPARQL endpoint contains the data set. VoID cannot express that a data set is contained a specific named graph. This can be specified with SPARQL 1.1. Service Description

VoID: Linksets metadata

- **Linkset:** a collection of RDF links between two RDF datasets
  - void:Linkset
  - void:target
  - void:linkPredicate

```
@PREFIX void:<http://rdfs.org/ns/void#>
@PREFIX owl:<http://www.w3.org/2002/07/owl#>

:DS1 a void:Dataset .
:DS2 a void:Dataset .
:DS1 void:subset :LS1 .
:LS1 a void:Linkset;
     void:linkPredicate owl:sameAs;
     void:target :DS1, :DS2 .
:LS2 a void:Linkset;
     void:linkPredicate owl:sameAs;
     void:target :DS1, :DS2 .
```

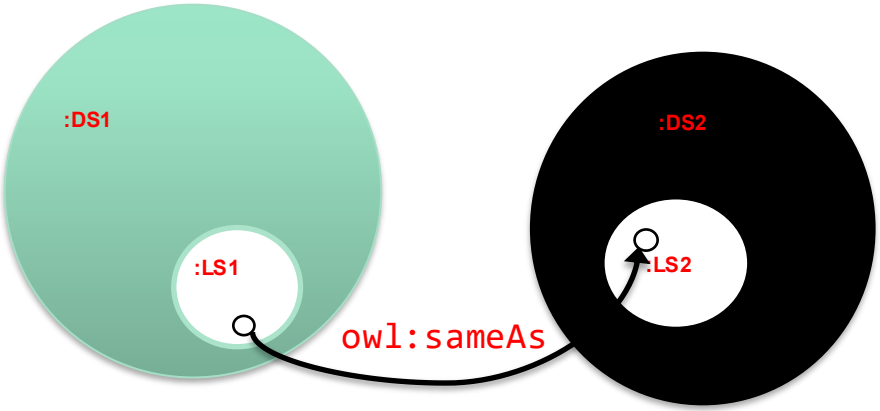
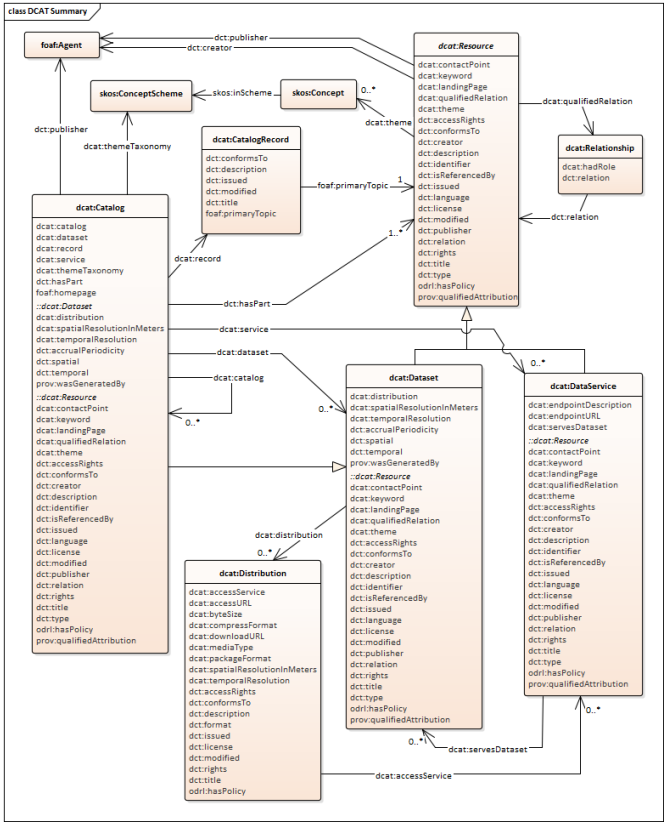


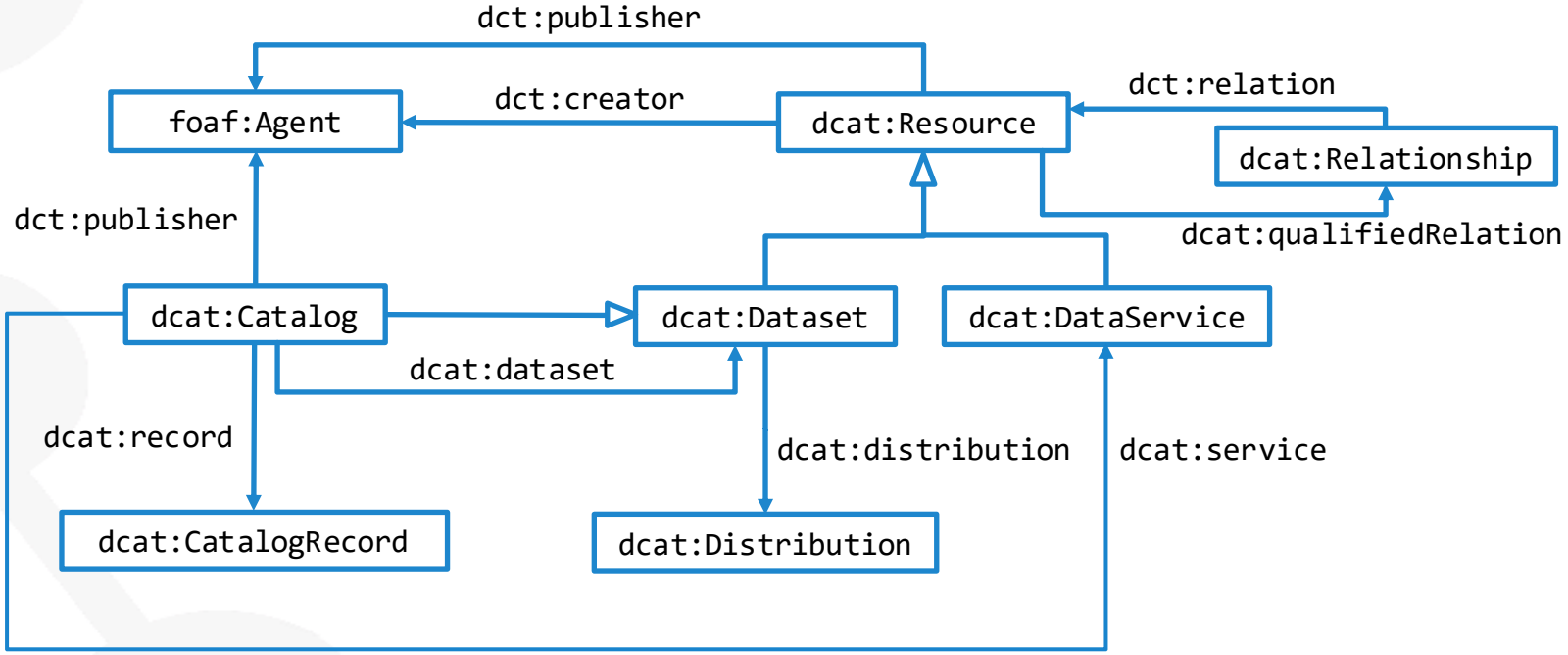
Image based on http://semanticweb.org/wiki/File:Void-linkset-conceptual.png

Data Catalog Vocabulary (DCAT)

- https://www.w3.org/TR/vocab-dcat
  - W3C recommendation (v1 in 2014; v2 in 2020; v3 in 2024)
  - Enables to describe datasets and data services in a catalogue
  - Facilitates interoperability between data catalogues published on the Web
  - Reuses predicates from well-known vocabularies such as Dublin Core Metadata Terms, FOAF, SKOS, among others
  - Has been extended to several domains: DCAT-AP, GeoDCAT-AP, StatDCAT-AP, CiteDCAT-AP, DCAT-AP-JRC, DCAT-AP_ES



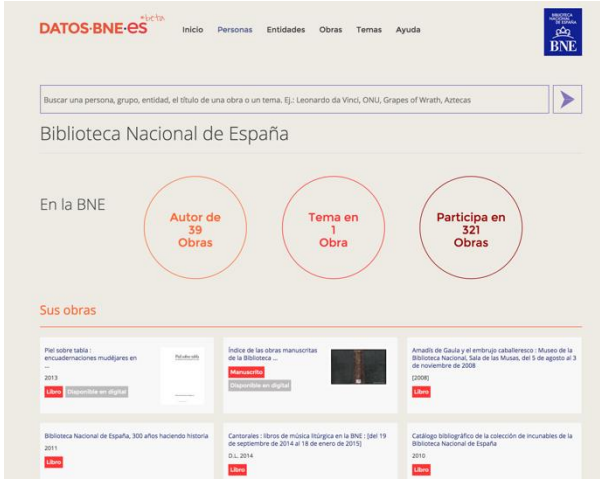
DCAT: Main concepts and relationships



Example: Metadata about datos.bne.es

- Combines DCAT and VoID, to include:
  - Licenses: Pointing to the rdflicense dataset
  - Provenance: Using PROV-O
  - Language information: Using lexvo.org
  - Access mechanisms: SPARQL endpoint, data dumps
- datos.bne.es is a catalogue
  - Composed of 7 (interlinked) datasets
  - With different distributions
    - Data dumps
    - SPARQL endpoint
    - Search API
- https://datos.bne.es/inicio.ttl

BNE: Spanish National Library



D. Vila-Suero, B. Villazón-Terrazas, and A. Gómez-Pérez. datos.bne.es: A library linked dataset. Semantic Web, 6(5):529–535

Ways to make Linked Data accessible on the Web

- Data dump
  - Places all dataset triples in one or more files in an RDF format
  - https://www.w3.org/wiki/DataSetRDFDumps
- SPARQL endpoint
  - Provides an endpoint on an HTTP network capable of receiving and processing SPARQL Protocol requests
  - Allows others execute queries on a server
  - https://www.w3.org/wiki/SparqlEndpoints
- Dereferencing URIs
  - Provides per-resource access to a dataset
  - Linked data documents
  - Read-write access (https://www.w3.org/TR/ldp/)

Tools to publish Linked Data: Virtuoso

- http://vos.openlinksw.com/owiki/wiki/VOS/
  - Open source and enterprise edition
  - HTML-based query service
  - Several SPARQL query result serialization formats
  - Faceted browsing tool
  - Configuration Steps:
    1. Download, install and start a local instance, pull a Docker container, or a preconfigured remote instance from the Amazon AWS Cloud.
    2. Download and install the Conductor (HTML-based DB Admin UI) or use the command line to change the default passwords for the *dba* and *dav* "super-user" accounts.
    3. Load RDF dataset.
  - Examples: DBpedia, Biblioteca Nacional de España, Datos Abiertos IGN, etc.

Tools to publish Linked Data: Blazegraph

- https://github.com/blazegraph/database
  - Open-source Java-based RDF graph database (natively supports SPARQL 1.1)
  - Provides a horizontally scaling storage and retrieval solution for very large volumes of RDF
  - Configuration steps:
    1. Download the last release *blazegraph.jar* file
    2. Run it by: *java -server -Xmx4g -jar blazegraph.jar*
    3. Go to *http://localhost:9999/blazegraph/*
    4. Load RDF dataset through the "Update" tab: *load <file://tmp/blazegraph/data.n3>*
  - Example: Wikidata
  - Acqui-hired by Amazon Web Services (Amazon Neptune)

Tools to publish linked data: GraphDB

- https://www.ontotext.com/products/graphdb/
  - Available in Free and Enterprise editions
  - Includes a user-friendly web interface
  - Supports inference
  - Configuration Steps:
    1. Download and run the GraphDB Desktop application for your operating system
    2. Create a new repository, providing an ID and configuring parameters like the desired ruleset for inference
    3. Import your RDF data by uploading files, providing a URL, or pasting raw data
    4. Use the SPARQL editor to query and explore the loaded data

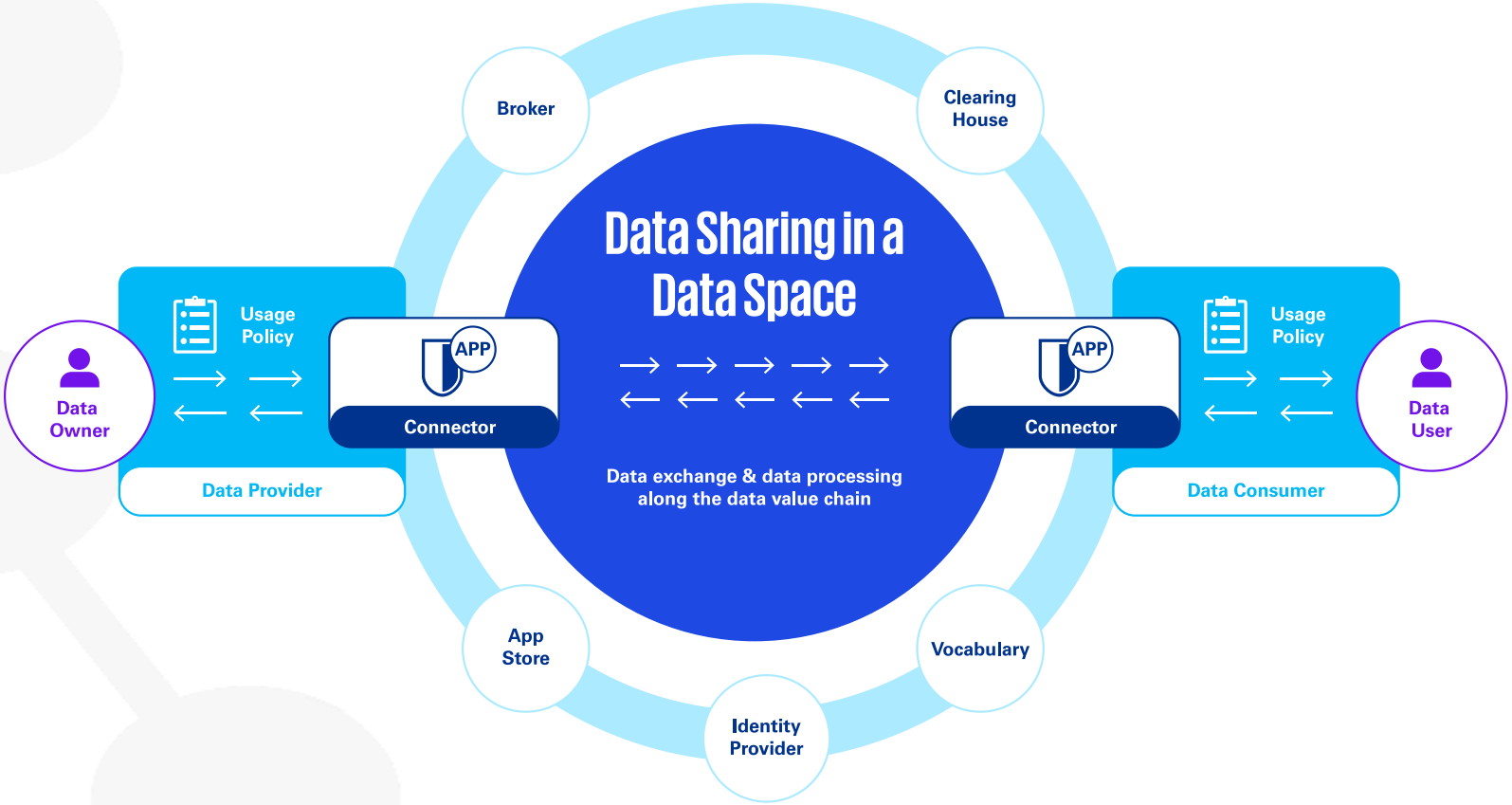
Inserting the dataset into Linked Data registries

- How do I get my dataset in the LOD cloud?
  - http://lod-cloud.net



- Validation
  - http://validator.lodcloud.net
- Registration in Datahub.io
  - http://esw.w3.org/TaskForces/CommunityProjects/LinkingOpenData/DataSets/CKANmetainformation

Inserting the dataset into a data space



https://kpmg.com/be/en/insights/technology/data-insights/demystifying-data-spaces-the-future-of-data-sharing.html

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Metadata for describing ontologies

- Consists of providing (machine-readable) metadata of ontologies
- This information allows for:
  - Generate human-readable documentation
  - Efficient and effective search of ontologies
- Several vocabularies for metadata description for ontologies and ontology terms

Metadata for describing ontologies

- Recommended:

Property name	Annotation property	Rationale
Title	dcterms:title	Understanding
Description	dcterms:description	Understanding
Creator	dcterms:creator	Provenance and attribution
Contributor	dcterms:contributor	Provenance and attribution
Creation date	dcterms:created	Provenance
Previous version	owl:priorVersion	Provenance and comparison
Namespace URI	vann:preferredNamespaceUri	Identifying the ontology
Version IRI	owl:versionIRI	Versioning
Prefix	vann:preferredNamespacePrefix	Identifying the ontology
Citation	dcterms:bibliographicCitation	Credit
License	dcterms:license	Usage conditions

Recommended for ontology terms

Property name	Annotation property	Rationale
Label	rdfs:label	Readibility
Definition	rdfs:comment	Understanding

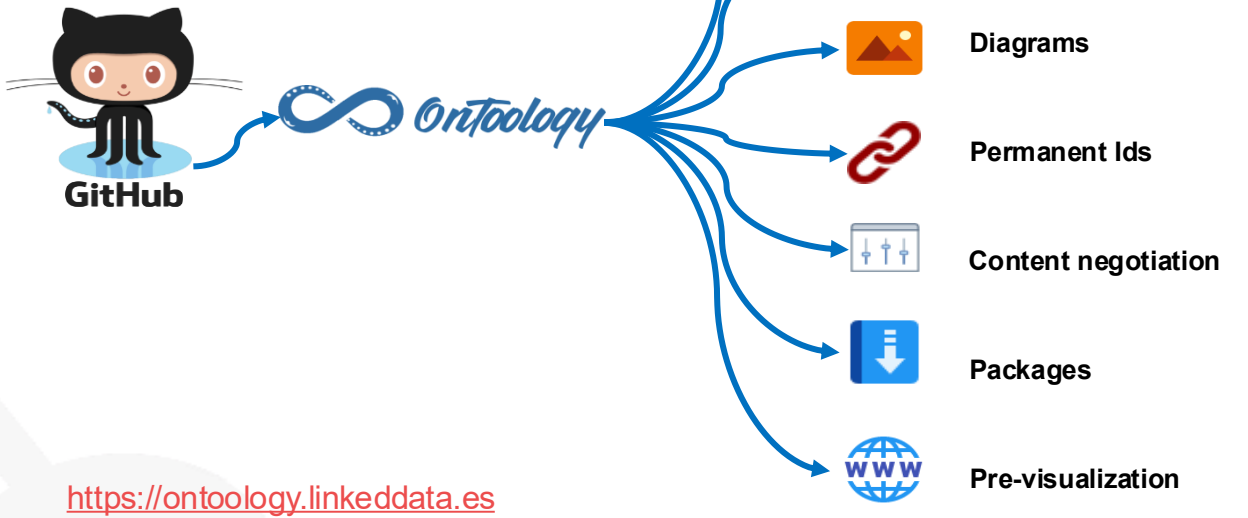
Garijo, D. and Poveda-Villalón M. “A checklist for complete vocabulary metadata.” http://dgarijo.github.io/Widoco/doc/bestPractices/index-en.html

Making the ontology available on the Web

- Consists of publishing the ontology on the Web following the best practices:
  - Proper documentation
  - Content negotiation
  - Evaluated
  - Permanent ID
- Tedious work …



Publishing ontologies



https://ontoology.linkeddata.es

Alobaid, A., Garijo, D., Poveda-Villalón, M., Santana-Pérez, I., Fernández-Izquierdo, A., & Corcho, O. "Automating ontology engineering support activities with OnToology". JWS (2018)

Making the ontology findable on the Web

- Register the ontology namespace prefix



https://prefix.cc/

- Register the ontology in metadata registries

Linked Open Vocabularies (LOV)

Ontobee

BioPortal

FAIRsharing

PIONERA
ontology hub

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Hands-on assignment 5

- **Goal**: to get familiar with the linking of RDF data with other datasets
- The students will have to take their selected dataset(s) and perform the following tasks:
  - **Data linking**
    - Identify those classes whose instances can be linked
    - Identify which data sets may contain instances for the previously-identified classes
    - Reconcile data with the identified datasets
  - **Transform into RDF**
    - Update the mappings and transform the data into RDF (using the Turtle syntax)
    - Write some SPARQL queries to check your data

Hands-on assignment (cont.)

- **Goal:** to get familiar with the publication of linked data and ontologies
- The students will have to take their knowledge graphs and perform the following tasks:
  - **Linked data publishing**
    - Publish your linked data with GraphDB
    - https://graphdb.ontotext.com/documentation/11.5/loading-data.html
    - Execute some SPARQL queries
  - **Ontology publishing (optional)**
    - Publish your ontology using Ontoology
    - https://ontoology.linkeddata.es/tutorial

Hands-on assignment 5 - Deliverables

For each dataset:

- A JSON file with the operations performed over the data for linking them
  - "***-with-links.json**" under a "lodrefine" directory
- A CSV file with the updated versions of the datasets
  - "***-with-links.csv**" under a "csv" directory
- An RML file with the updated versions of the mappings
  - "***-with-links.rml**" under a "mappings" directory
- An RDF file in the Turtle syntax with the data transformed into RDF
  - *"***-with-links.ttl**"* under an "rdf" directory
- A SPARQL file with queries to verify your links
  - "**queries-with-links.sparql**" under an "rdf" directory

Assignment deadline: October, 24th

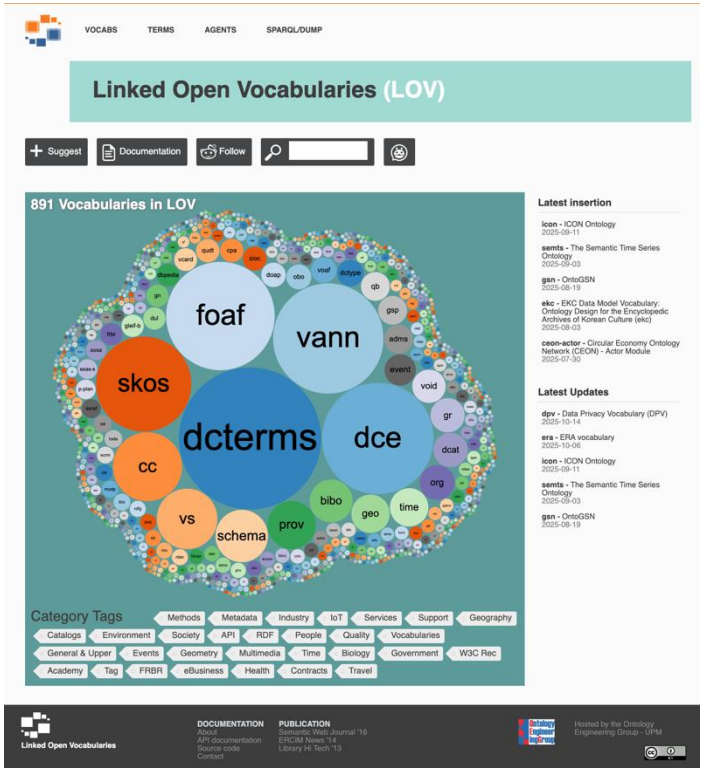
Hands-on assignment 5 – Self-assessment

- **Every RDF file:**
  - Contains at least one *owl:sameAs* property
- **Every owl:sameAs property:**
  - Links a resource in the dataset with another resource in an external dataset
- **The SPARQL file:**
  - Includes at least 1 query
  - Contains queries that retrieve the data that would be needed in the application
- **Every SPARQL query:**
  - Makes use of the ontology
  - Returns a non-empty result
  - Makes use of the *owl:sameAs* links

- To get familiar with the LOV ontology catalogue
  - https://lov.linkeddata.es/dataset/lov/

- To take pre-class test 6





Questions?



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